WATERSHED SEGMENTATION

STUDY NOTES

Luke Camilleri, Jake Desira, Kurt Abela, Tristin Bezzina, Nathan Vella

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1. Introduction

Image segmentation is one of the most fundamental problems in computer vision: given a digital image, partition it into meaningful regions that correspond to objects, surfaces, or structures of interest. Applications span an enormous range, from delineating tumour boundaries in medical scans, to separating touching cells in histology, to identifying land-cover classes in satellite imagery. Despite decades of research, segmentation remains a non-trivial task, because the notion of a "meaningful region" is inherently dependent on context, scale, and the nature of the underlying image.

Among the classical approaches to segmentation, the watershed transform occupies a distinctive position. Rather than operating on the image as a flat signal, watershed treats it as a three-dimensional landscape: pixel intensities become elevations, and the problem of finding boundaries is recast as one of hydrology. The metaphor is immediate and powerful. Imagine rainfall falling uniformly across a hilly terrain. Water flows downhill, collecting in the lowest points of each valley. Over time, it fills each valley to form a lake, and where the floodwaters from two adjacent lakes would meet, a dam is built. The collection of all such dams traces out the ridgelines of the terrain, and in the image domain, those ridgelines correspond precisely to the boundaries between image regions.

This intuition was formalised for digital image processing by Beucher and colleagues at the Centre de Morphologie Mathématique (CMM), Mines Paris, building on the mathematical morphology framework developed by Serra and Matheron. Beucher's formulation established the conceptual and mathematical vocabulary, catchment basins, watershed lines, regional minima, that remains standard in the field today. Crucially, the approach is grounded not in heuristics but in a well-defined topological framework, giving it both theoretical rigour and geometric interpretability.

These notes provide a comprehensive treatment of watershed segmentation, bridging conceptual intuition and technical detail. Section 2 formalises the topographical metaphor and establishes the mathematical foundations. Section 3 introduces the gradient-based watershed, the most widely used practical formulation, and presents a step-by-step worked numerical example. Section 4 addresses the problem of over segmentation and the use of markers to constrain the transformation. Section 5 examines the principal failure modes of watershed segmentation. Section 6 surveys alternative segmentation methods and contextualises watershed within the broader landscape of techniques. Section 7 presents real-world applications. The notes conclude with a Quick Reference cheat sheet and a curated list of key papers.

2. Topographical Metaphor

2.1 From Image to Landscape

The central insight of watershed segmentation is a change of representation: a greyscale image is not merely a two-dimensional array of intensity values, but a topographic surface embedded in three-dimensional space. Formally, let $f : D \subseteq \mathbb{Z}^2 \to \mathbb{Z}$ be a digital greyscale image defined on a finite rectangular domain D, where f(x, y) denotes the intensity of the pixel at position (x, y). The topographic surface S^p associated with f is the graph:

$$S^p = \{ (x, y, z) \in \mathbb{Z}^2 * \mathbb{Z} : z = f(x, y) \}$$

embedded in the three-dimensional lattice $\mathbb{Z}^3$. Under this mapping, each pixel (x, y) corresponds to a point on a surface at height f(x, y). Bright pixels. those with high intensity, correspond to elevated terrain; dark pixels correspond to valleys or depressions. This reinterpretation turns the problem of segmenting an image into the problem of partitioning a terrain into drainage regions, a question studied in geomorphology long before its application to image analysis.

The analogy is not merely aesthetic. In natural landscapes, the drainage structure of a terrain is determined entirely by its topography: water always flows in the direction of steepest descent, and every point on the surface belongs to the catchment area of exactly one local minimum, a valley bottom, unless it lies precisely on a ridgeline separating two drainage basins. The watershed transform applies this same logic to the image surface, exploiting the geometric structure of intensity gradients to produce a principled partition of the image domain.

2.2 Key Definitions

Three concepts are central to the topographical formulation, each with a precise mathematical definition.

A regional minimum of f is a connected set $M \subseteq D$ such that $f(p) = c$ for all $p \in M$ (M is flat at height c), and $f(q) > c$ for every pixel $q \in D$ that is adjacent to M but not in M. In topographic terms, a regional minimum is a valley floor from which there is no downhill path leading out: any step away from M moves to higher ground. The set of all regional minima of f is denoted M(f).

The catchment basin $CB(m_i)$ of a regional minimum m_i is the set of all pixels $p \in D$ such that a drop of water placed at p and following the path of steepest descent would eventually reach m_i. Formally,

$$CB(m_i) = \{ p \in D : \text{the steepest-descent path from } p \text{ terminates at } m_i \}$$

The catchment basins partition the image domain D into disjoint regions, each associated with one regional minimum. In a natural image, these basins typically correspond to coherent regions of relatively uniform intensity, flat or smoothly varying areas separated by intensity edges.

The watershed line (or watershed transform) W(f) is the complement of the union of all catchment basins:

$$W(f) = D \setminus \bigcup_i CB(m_i)$$

Geometrically, W(f) consists of those pixels that lie on the ridges of the surface S^p, points from which water might flow to two or more distinct regional minima. These ridgelines correspond, in the image domain, to the boundaries between regions of differing intensity. The watershed line is therefore a closed, connected set of boundary pixels that divides D into as many regions as there are regional minima.

2.3 The Flooding Simulation

Beucher formalised the watershed transform using the metaphor of progressive flooding from below. Consider the topographic surface S^p submerged upside-down in a body of water, or equivalently, imagine water rising uniformly from the lowest points of the landscape upward. At a given water level h, the flooded region is

$$T_h(f) = \{ p \in D : f(p) \leq h \}$$

As h increases from the global minimum to the global maximum of f, water first floods the deepest regional minima, forming isolated lakes. As the level rises, each lake expands outward. The critical event occurs when two lakes from different regional minima are about to merge: at precisely this level, a dam is constructed along the boundary between them. This dam, a single-pixel-wide ridge, is added to the watershed line. The flooding continues, with dams preventing the merging of basins, until the entire surface is submerged.

The flooding simulation is not merely a visualisation device. It provides a constructive definition of the watershed transform and directly motivates efficient algorithmic implementations.

2.4 Beucher's Contribution and Historical Context

The application of the watershed concept to image segmentation was pioneered by Beucher and Lantuéjoul at the CMM in the late 1970s, within the framework of mathematical morphology. Mathematical morphology, introduced by Matheron and Serra, provides a set-theoretic language for analysing the geometry of images, emphasising shape and structure over pixel-level statistics. The watershed transform fit naturally into this framework: like other morphological operators, it operates on the topological structure of the image rather than on individual pixel values in isolation.

Beucher's key contribution was to establish that the topographic metaphor is not merely an analogy but a mathematically precise correspondence. By embedding the image as a surface in $\mathbb{Z}^3$ and importing concepts from geomorphology, regional minima, drainage basins, divides, he provided image segmentation with a rigorous geometric foundation. This had two important consequences.

First, it made the watershed transform amenable to formal analysis: properties such as the thinness of the watershed line and the completeness of the basin partition could be proved, not merely observed empirically. Second, it suggested a natural generalisation: any scalar function defined on a discrete domain can be treated as a topographic surface and subjected to the same analysis, including, crucially, the gradient magnitude of a greyscale image, which forms the basis for the gradient-based watershed discussed in Section 3.

3. Gradient-Based Watershed

3.1 Motivation: Why Not Apply Watershed Directly to Intensity?

A natural first attempt at watershed segmentation would be to apply the transform directly to the greyscale image f. Under this approach, the regional minima of f, the darkest points in each region, would seed catchment basins, and the watershed lines would trace boundaries between them. In practice, however, this approach fails for most natural images. Objects in an image are not defined by their intensity minima; they are defined by the transitions between regions. A dark cell against a bright background, for instance, has its lowest intensity values deep within its interior, far from the boundary that we actually want to detect. Applying watershed to f would place the basin seeds inside the cell rather than at its edges, and the resulting watershed lines would bear no useful relationship to object boundaries.

The solution, formalised by Beucher and Meyer, is to apply the watershed transform not to the intensity image f itself, but to its gradient magnitude g(f). The gradient is a measure of local intensity variation: it is high wherever the image changes rapidly, precisely at the transitions between objects, and low wherever the image is smooth and homogeneous. The regional minima of g(f) therefore correspond to regions of uniform intensity, which is exactly where object interiors lie. The watershed lines of g(f), the ridges of the gradient surface, correspond to the sharp transitions between regions, which is exactly where object boundaries lie. This reframing transforms watershed segmentation from a method that detects intensity minima into one that detects edges.

Bai et al. (2017) note that this is precisely how the classical watershed transform is standardly deployed in practice: it is typically applied to the image gradient so that catchment basins correspond to homogeneous regions of the original image. Their own work, which replaces the hand-crafted gradient with a learned energy map from a

convolutional neural network, is premised on this same principle, validating the gradient-based approach as the correct conceptual foundation even as it extends it with modern machinery.

3.2 The Morphological Gradient

While several gradient operators can be used, including the Sobel, Prewitt, and Canny edge detectors, the morphological gradient is particularly well suited to watershed segmentation because it is defined within the same mathematical morphology framework as the transform itself. As defined by Beucher and Meyer, the morphological gradient of f with respect to a structuring element B is:

$$g(f) = (f \ [+] \ B) - (f \ [-] \ B)$$

where [+] and [-] denote the morphological dilation and erosion operators respectively. For a pixel at position x, these are defined as:

$$(f \ [+] \ B)(x) = \max\{ f(y) : y \text{ in } Bx \}$$

$$(f \ [-] \ B)(x) = \min\{ f(y) : y \text{ in } Bx \}$$

where Bx = { x + b : b in B } is the structuring element B translated to position x. The dilation at x takes the maximum intensity in a local neighbourhood defined by B, and the erosion takes the minimum. The morphological gradient is their difference: a measure of the spread of intensity values within that neighbourhood.

The structuring element B is typically a small symmetric shape: a 3x3 square or cross in 2D, or a symmetric interval of length 3 in 1D. The choice of B controls the spatial scale at which intensity transitions are detected. A larger B will smooth over fine details and emphasise coarser boundaries, while a smaller B responds to sharper, finer-scale transitions.

The morphological gradient satisfies g(f)(x) >= 0 everywhere, since the maximum of any set is always at least as large as its minimum. It equals zero on any region where f is constant within the B-neighbourhood (a perfectly flat area contributes zero spread). It is large wherever f changes rapidly, such as at the boundary between a bright and a dark region. These properties make g(f) a natural topographic surface for watershed segmentation: its regional minima sit inside homogeneous object regions, and its ridges sit at object boundaries.

3.3 The Flooding Algorithm on g(f)

With g(f) computed, the gradient-based watershed proceeds by applying the immersion flooding process to the gradient surface. In practice, the standard implementation uses a priority queue: all pixels are sorted in ascending order of their gradient value and processed from lowest to highest, with basin labels propagating outward level by level.

For each pixel removed from the queue, its labelled neighbours are examined. If all labelled neighbours belong to the same catchment basin, the pixel is assigned to that basin. If labelled neighbours from two or more distinct basins are found, the pixel is marked as a watershed point WSHED. If the pixel has no labelled neighbours at all, it is the seed of a new catchment basin: it receives a new unique label and propagates that label to its unlabelled neighbours at the same gradient level via a FIFO sub-queue. The process continues until all pixels have been assigned a label or marked as WSHED, producing a complete partition of the domain.

3.4 Worked Numerical Example

Consider the following one-dimensional signal of seven pixels p1 through p7, representing a cross-section through an image with two dark objects separated by a bright peak:

Pixel	p1	p2	p3	p4	p5	p6	p7
f(pi)	3	1	2	8	2	1	3

The signal has two intensity minima (p2 = 1 and p6 = 1), representing the interiors of two dark objects, and a central peak (p4 = 8), representing a bright boundary region between them.

3.4.1: Compute the morphological gradient

using B = {−1, 0, +1} with mirror-padding at the boundaries:

$$g(pi) = \max\{ f(pi-1), f(pi), f(pi+1) \} - \min\{ f(pi-1), f(pi), f(pi+1) \}$$

Pixel	Neighbours	Dilation	Erosion	g(pi)	Regional min?
p1	3, 3, 1	3	1	2	Yes
p2	3, 1, 2	3	1	2	Yes
p3	1, 2, 8	8	1	7	-
p4	2, 8, 2	8	2	6	Yes
p5	8, 2, 1	8	1	7	-
p6	2, 1, 3	3	1	2	Yes
p7	1, 3, 3	3	1	2	Yes

The resulting gradient signal is g = [2, 2, 7, 6, 7, 2, 2]. There are three regional minima: {p1, p2} at g = 2 (left dark object), {p4} at g = 6 (bright peak, locally flat at its apex), and {p6, p7} at g = 2 (right dark object).

3.4.2: Sort and flood, pixels processed in ascending order of g(pi):

Step	h	Pixel	Decision	State [p1…p7]

1	2	p1	No labelled neighbours → seed Basin B1	[B1, --, --, --, --, --, --]
2	2	p2	Neighbour p1 = B1 → assign B1	[B1, B1, --, --, --, --, --]
3	2	p3	No labelled neighbours → seed Basin B2	[B1, B1, --, --, --, B2, --]
4	2	p4	Neighbour p6 = B2 → assign B2	[B1, B1, --, --, --, B2, B2]
5	6	p5	No labelled neighbours → seed Basin B3	[B1, B1, --, B3, --, B2, B2]
6	7	p6	Neighbours p2=B1, p4=B3 → two basins → WSHED	[B1, B1, W, B3, --, B2, B2]
7	7	p7	Neighbours p4=B3, p6=B2 → two basins → WSHED	[B1, B1, W, B3, W, B2, B2]

3.4.3: Results

The final segmentation gives Basin B1 = {p1, p2}, Basin B3 = {p4}, Basin B2 = {p6, p7}, with watershed lines at W = {p3, p5}. The watershed lines fall precisely at the steepest transitions in the original signal, confirming that gradient-based watershed correctly localises boundaries at intensity edges.

3.5 Application: The Bubble Segmentation Example

Beucher and Meyer illustrate the failure of direct intensity watershed with a practical example: a greyscale image of circular bubbles. Applying the watershed transform directly to f produces poor results, because the regional minima of f lie deep within the bright interiors of the bubbles. Applying the morphological gradient g(f) and then running the watershed produces a qualitatively different result. The gradient is high along the bubble walls, at the sharp transitions between bright bubble interiors and the darker background, and low both inside the bubbles and in the uniform background. The watershed lines of g(f) therefore trace the bubble contours accurately, demonstrating that the gradient-based approach is both necessary and sufficient for this class of problem.

This example encapsulates the core practical lesson: the method succeeds when object boundaries correspond to sharp local intensity transitions, which covers a large and important class of problems in microscopy, industrial inspection, and medical imaging.

4. Markers and Over Segmentation

4.1 The Over-Segmentation Problem

The gradient-based watershed transform is theoretically complete but practically problematic: it treats every local minimum of the gradient surface as the seed for a

distinct catchment basin. Real images contain hundreds or thousands of spurious minima caused by noise, texture variation, and quantisation artefacts. The result is over-segmentation, a partition containing far more regions than meaningful objects.

Formally, if gradient image g has k regional minima, the watershed transform produces exactly k catchment basins. For a typical 256 × 256 greyscale image, k may exceed 10,000, rendering the output analytically useless without further processing.

4.2 What Is a Marker?

A marker is a connected set of pixels used to identify a region of interest before flooding begins. Two classes exist. Internal markers are blobs of pixels located inside each foreground object, one per object to be segmented, seeding that object's catchment basin. External markers are pixels known to belong to the background, forming a single additional basin that absorbs all unmarked space. Markers may be placed manually by an expert or computed automatically using morphological operations such as opening-by-reconstruction and closing-by-reconstruction (Kaur & Jindal, 2011).

4.3 Marker Imposition

Once foreground markers M_{fg} and background markers M_{bg} are identified, the gradient image g is modified so that its only regional minima coincide with those marker locations. This is achieved through geodesic reconstruction, a morphological fill that converts all unmarked minima into non-minimum plateaux, preventing them from generating new catchment basins. The modified gradient g^* satisfies:

$$\text{Regional_Minima}(g^*) = M_{fg} \cup M_{bg}$$

Watershed is then applied to g^*, producing exactly $|M_{fg}| + 1$ catchment basins: one per foreground object plus one background region.

4.4 Five-Step Procedure

The standard marker-controlled pipeline proceeds as follows (Kaur & Jindal, 2011):

1. Compute the segmentation function (typically the morphological gradient g).
2. Compute foreground markers connected blobs inside each foreground object.
3. Compute background markers, pixels confirmed not to belong to any object.
4. Impose markers on g via geodesic reconstruction to yield modified gradient g^*.
5. Apply the watershed transform to g^*.

4.5 Practical Effect

On a benchmark lung CT slice, Kaur and Jindal (2011) show that plain watershed produces a heavily fragmented partition, while marker-controlled watershed reduces this

to anatomically meaningful structures, cleanly delineating the lung boundary with no spurious interior splits. The key insight is simple: the number of output regions is controlled by the analyst, not by the noise floor of the image.

5. Failure Modes

Despite its theoretical rigour, gradient-based watershed exhibits well-documented failure modes that limit its reliability in practice. Gilani et al. (2025) conduct an empirical reliability audit of marker-free watershed segmentation on dermoscopic images, reporting catastrophic failure rates, defined as cases where the Dice overlap score falls below 0.1, of between 20% and 33% even under clean imaging conditions, a figure that worsens substantially under noise, blur, and contrast degradation.

Over-segmentation.

The most cited failure mode arises when the gradient image g(f) contains too many regional minima. Noise, fine texture, and minor intensity fluctuations each introduce spurious local minima, and each minimum seeds its own catchment basin. The result is a fragmented partition containing many small, meaningless regions where only a few objects exist. Bai et al. (2017) identify this as the central limitation of classical watershed and the primary motivation for replacing the hand-crafted gradient with a learned energy surface.

Boundary leakage and under-segmentation.

Where the gradient is weak, at low-contrast boundaries where two regions transition gradually, the ridge between two potential basins is too shallow to form a stable watershed line. The flooding process merges basins across the intended boundary, producing a single region where two distinct objects exist. Gilani et al. identify this as the most clinically dangerous failure mode, since a missed boundary in a medical context corresponds directly to a missed lesion.

Noise and artifact sensitivity.

Gradient operators amplify high-frequency noise: small intensity perturbations produce large local gradient values, introducing spurious minima throughout g(f). Gilani et al. demonstrate that under controlled noise and blur perturbations, watershed catastrophic failure rates increase to levels that make the method unreliable without dedicated preprocessing or marker-based constraints.

6. Alternatives to Watershed Segmentation

When watershed's gradient-dependent flooding produces unreliable results, two principal alternative paradigms offer distinct approaches to the segmentation problem.

Graph Cuts reframe segmentation as a global energy minimisation problem. Peng and Liu (2010) define the energy function as:

$$E = \Sigma V_{p,q} + \Sigma D_p$$

where $V_{p,q}$ is the penalty for a discontinuity between neighbouring pixels p and q, and D_p is the data energy measuring how well a label fits a pixel given observed intensity. The minimum of this energy function is found using a max flow/min cut algorithm, which finds the global optimum across the entire image simultaneously — in direct contrast to watershed's greedy, local flooding process. Peng and Liu (2010) demonstrate that using watershed to pre-segment the image into regions, and then applying graph cut over those regions rather than individual pixels, reduces computation time significantly: on a 256×256 image, running time falls from 2.44 seconds to 0.63 seconds while producing more accurate object boundaries than either method alone. The key advantage of graph cut over plain watershed is robustness — by encoding both boundary and region information into a single energy function, it avoids the over-segmentation that arises from watershed's reliance on gradient minima alone.

Active Contours take a fundamentally different approach: rather than flooding from minima, a curve is initialised near a boundary and iteratively deformed until it settles on the object edge by minimising an energy functional. Baswaraj et al. (2012) identify this as a key advantage over gradient-based methods — active contours always produce sub-regions with continuous boundaries, whereas kernel-based edge detectors such as Sobel and Canny regularly produce discontinuous ones. Two implementations exist: snakes, which explicitly shift a set of predefined contour points toward a local energy minimum, and level sets, introduced by Osher and Sethian, which represent the contour implicitly as the zero level of a higher-dimensional function and can therefore handle topology changes such as splitting and merging that snakes cannot. Baswaraj et al. (2012) further distinguish edge-based active contours — which, like watershed, rely on the image gradient and inherit its sensitivity to noise and local minima — from region-based active contours, which use global energy minimisation over intensity statistics inside each region and therefore require no prior knowledge of contour placement.

Both alternatives carry trade-offs: graph cut requires user-defined seed regions and is computationally heavier than watershed's O(n) flooding, while active contours require careful initialisation and iterate until convergence. Watershed remains preferable when no labelled seeds are available and images contain strong, well-defined gradient ridges.

7. Real-World Applications

Watershed segmentation has been applied across a wide range of domains. In medical imaging, marker-controlled watershed is used to delineate anatomical structures from

CT and MRI data; Kaur and Jindal (2011) demonstrate clean lung lobe segmentation from DICOM images using the marker-controlled pipeline. In dermatology, Gilani et al. (2025) apply watershed to dermoscopic lesion boundary detection, using their results to establish reliability benchmarks for clinical deployment. In industrial vision, Niu et al. (2025) deploy an adaptive watershed framework for real-time particle size measurement in dynamic fluid flows, where speed and training-free operation are essential constraints. In computer vision research, Bai and Urtasun (2017) use the watershed framework as the backbone of an instance segmentation system for autonomous driving scenes, replacing the hand-crafted gradient with a learned energy map to handle complex, cluttered environments. Across all these domains, the common thread is the same: watershed succeeds when object boundaries correspond to high-gradient transitions, and fails, or requires augmentation, when they do not.

Quick reference

Core formula: morphological gradient: $g(f) = (f \oplus B) - (f \ominus B)$

Flooding decision rule (per pixel p removed from queue):

- All labelled neighbours same basin → assign p to that basin
- Labelled neighbours from ≥ 2 basins → mark p as WSHED
- No labelled neighbours → seed new basin, assign new label

Term	**Definition**
Regional minimum	Connected flat region with no lower adjacent pixel
Catchment basin CB(m_i)	All pixels whose steepest-descent path leads to m_i
Watershed line W(f)	Complement of all catchment basins; the boundary set
Morphological gradient	g(f) = dilation − erosion; measures local intensity spread
Structuring element B	Neighbourhood shape used in dilation/erosion
Marker	Connected pixel set seeding a catchment basin
Geodesic reconstruction	Morphological fill that eliminates unmarked minima
Over-segmentation	Too many basins due to spurious gradient minima
Catastrophic failure	Dice score < 0.1; complete segmentation breakdown

References:

Beucher, S. (n.d.). Watershed, Hierarchical Segmentation and Waterfall Algorithm. Centre de Morphologie Mathématique, Mines Paris. Available at: Article link

The foundational paper establishing watershed segmentation within the mathematical morphology framework. Beucher formalises the topographic metaphor — greyscale image as terrain, pixel intensity as elevation — and defines regional minima, catchment basins, and watershed lines with mathematical precision. The flooding simulation is introduced as both a conceptual device and an algorithmic basis. The paper also presents the waterfall algorithm, a hierarchical extension that produces multi-scale segmentations by iteratively merging the shallowest watershed lines. Essential reading for understanding the theoretical foundations of all watershed-based methods.

Beucher, S. & Meyer, F. (1993). The Morphological Approach to Segmentation: The Watershed Transformation. In E. R. Dougherty (Ed.), Mathematical Morphology in Image Processing (pp. 433–481). Marcel Dekker.Article Link

This chapter establishes the gradient-based watershed as the practical standard, arguing that the transform should be applied to the morphological gradient $g(f) = (f \oplus B) - (f \ominus B)$ rather than to the raw intensity image. The authors define dilation and erosion formally, prove the gradient's key properties, and demonstrate the approach on bubble segmentation. The priority-queue flooding algorithm is presented and its O(n) complexity established. This work is the direct source of the morphological gradient formulation and flooding algorithm used throughout modern implementations.

Bai, M. & Urtasun, R. (2017). Deep Watershed Transform for Instance Segmentation. Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), pp. 5221–5229. Article Link

Proposes replacing the hand-crafted morphological gradient with a learned energy map produced by a convolutional neural network, trained to predict direction-aware distance transforms that peak at object centres. The watershed cut is then applied to this learned surface rather than the image gradient, dramatically reducing over-segmentation on cluttered scenes. Evaluated on the CVPR instance segmentation benchmark, the method outperforms classical watershed and competitive proposal-based approaches. The paper validates the watershed framework as a sound conceptual foundation while demonstrating that its main limitation, sensitivity to the gradient surface, can be overcome with learned representations.

Gilani, Z. F., Mansoor, I., & Zhao, H. (2025). Reliability Auditing of Medical Image Segmentation under Signal Variability: The Catastrophic Failure Rate Framework. SSRN Preprint. Article Link

Introduces the Catastrophic Failure Rate (CFR) metric — defined as the proportion of test cases where Dice overlap falls below 0.1 — as a reliability measure for segmentation methods under real-world signal degradation. Applied to marker-free watershed on dermoscopic images, the study reports CFRs of 20–33% under clean conditions, rising sharply under noise, blur, and contrast perturbation. The paper identifies boundary leakage at low-contrast edges as the dominant failure mode and argues that reliability auditing should be a standard component of any clinical deployment evaluation, alongside mean accuracy metrics.

Kaur, M. & Jindal, G. (2011). Medical Image Segmentation using Marker Controlled Watershed Transformation. *International Journal of Computer Science and Technology*, 2(4), 548–551. Article Link

Peng, Y. & Liu, R. (2010). Object Segmentation Based on Watershed and Graph Cut. *2010 3rd International Congress on Image and Signal Processing (CISP2010)*, pp. 1431–1435. Article Link

Baswaraj, D., Govardhan, A. & Premchand, P. (2012). Active Contours and Image Segmentation: The Current State of the Art. *Global Journal of Computer Science and Technology*, 12(11). Article Link